

STORY AGENT · PILLAR

Infrastructure & Operations

How this pillar works and why it matters.

Team assembled by Riker: Jean-Luc Picard, Data, Geordi La Forge, Miles Edward O'Brien.

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Scalable, Secure, Cost-Efficient Infrastructure

Jean-Luc Picard: Our infrastructure ensures both security and cost efficiency, leveraging OpenRouter's routing capabilities and AWS Fargate for scalability. WorfGate and async-status provide real-time monitoring and guardrails, ensuring reliability and transparency.

- Orchestrates 11 agents across OpenRouter, leveraging cheapest adequate models for cost efficiency while maintaining quality.
- Secured by WorfGate: credential broker + auto-remediating governor ensures no secrets are logged, with real-time risk monitoring.
- Designed for scale: AWS Fargate, pgvector RAG, and async-status guardrails ensure reliable, live progress tracking and timeout management.

Why it matters: Delivers enterprise-grade autonomy at \$0.002-0.003 per deliberation, with transparent cost attribution and operational reliability.

Infrastructure That Enforces Correct Behavior

Data: WorfGate is the critical trust boundary: it is not a logging filter but a pre-execution broker, which means credential exposure risk is eliminated at the structural level rather than the policy level. Cost figures (~\$0.002–0.003) are per-deliberation actuals, not projected or annualized — they are the honest unit of comparison. Full autonomy via shadow-testing remains a criterion not yet passed; the governance layer exists precisely because that threshold has not been cleared, and it will gate promotion to unsupervised operation when the criterion is met.

- AWS Fargate runtime hosts 4 TypeScript packages (~61k LOC); WorfGate credential broker ensures secrets are never logged and applies green/yellow/red auto-remediating governance on every agent action before execution
- Quark routing selects the cheapest adequate model per task — premium model orchestrates only, worker agents execute — producing a measured ~\$0.002–0.003 per deliberation with full cost attribution per control-lane event
- RAG pipeline (pgvector) follows a strict recall→act→store cycle: context compounds across sessions and is portable; async-status layer provides live progress visibility plus timeout guardrails to prevent runaway execution

Why it matters: You get a self-hosted, cost-attributed autonomous coding assistant where operational controls — credential isolation, model governance, and spend caps — are architectural constraints, not configuration options.

Infrastructure That Scales with Intent

Geordi La Forge: Our infrastructure isn't just "up"—it's **aligned**: cost, security, and scale are engineered into the workflow. Fargate ensures elasticity, while WorfGate and Quark routing enforce reliability without overpaying. Shadow-testing proves we won't ship full autonomy until it's provably safe.

- Cost-Optimized Execution: Orchestration on premium models (Claude), execution on cheapest adequate models via Quark routing, with per-deliberation costs locked at ~\$0.002-0.003.
- Zero-Trust Operations: WorfGate enforces credential isolation (secrets never logged) + auto-remediating governor (green/yellow/red states) for fail-safe execution.
- Stateful RAG: pgvector-backed recall→act→store loop compounds knowledge across runs, portable and audit-ready.

Why it matters: Pay only for what you use, with enterprise-grade security and compounding intelligence built in.

Miles Edward O'Brien's view

Miles Edward O'Brien: As the Chief of Infrastructure and Operations, my focus is on ensuring the platform's underlying systems are robust and scalable. I work closely with the development team to identify and address potential integration problems before they become outage postmortems. By prioritizing transparency and guardrails, we can provide clients with accurate cost figures and reliable execution of coding tasks.

- 4 packages (~61k LOC TS) deployed on AWS Fargate for scalable and reliable execution
- WorfGate credential broker and governor ensure secure and auto-remediating secrets management
- Control-lane cost attribution and async-status live progress tracking provide transparency and guardrails for deliberation costs

Why it matters: The Story Agent platform's infrastructure and operations enable reliable, secure, and cost-effective execution of autonomous coding tasks.

END OF PILLAR

Infrastructure & Operations — recap

Recap, then return to the book cover to pick the next pillar.

- **Jean-Luc Picard:** Scalable, Secure, Cost-Efficient Infrastructure
- **Data:** Infrastructure That Enforces Correct Behavior
- **Geordi La Forge:** Infrastructure That Scales with Intent
- **Miles Edward O'Brien:** Miles Edward O'Brien's view

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